



Univerzita Hradec Králové
Filozofická fakulta

E-learningový kurz

Modern quantitative methods
and shape analysis in archaeology



EVROPSKÁ UNIE
Evropské strukturální a investiční fondy
Operační program Výzkum, vývoj a vzdělávání



MINISTERSTVO ŠKOLSTVÍ,
MLÁDEŽE A TĚLOVÝCHOVY

Tento materiál vznikl v rámci realizace projektu
Strategický rozvoj Univerzity Hradec Králové,
reg. č. CZ.02.2.69/0.0/0.0/16_015/0002427.

2D landmark analyses

Practice



EVROPSKÁ UNIE
Evropské strukturální a investiční fondy
Operační program Výzkum, vývoj a vzdělávání



Hadrian
(117-138 AD)



Antonius Pius
(138-191 AD)

Quantitative comparison of two sets of coins

Questions:

- Are coins/faces of Hadrian and his successor Antonius Pius different?
- Can we visualise these differences?

Dataset

- 22 coins of Hadrian
- 15 coins of Antonius Pius

Procedure

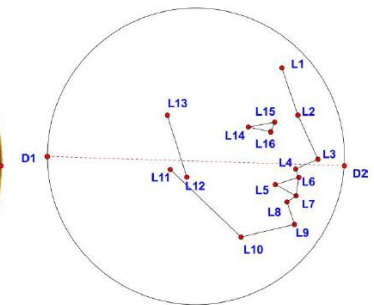
- Digitize landmarks
- Perform General Procrustes Analysis (GPA)
- Perform Relative Warp Analysis (RWA)
- Perform Thin-plate splines (TPS)
- Interpretation of results



Hadrian
(117-138 AD)



Antonius Pius
(138-191 AD)



Software

- tps: <https://life.bio.sunysb.edu/morph/>
- PAST: <https://folk.uio.no/ohammer/past/>



Hadrian
(117-138 AD)



Antonius Pius
(138-191 AD)



tps Utility program
Version 1.78

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tpsDig2
Version 2.31

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Relative warps
Version 1.70

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tpsSuper
Version: 2.05

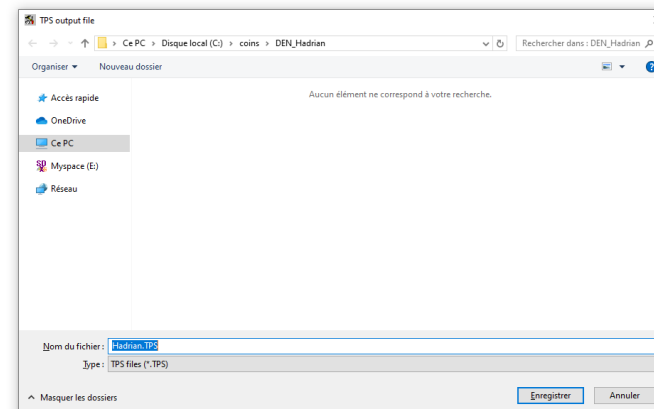
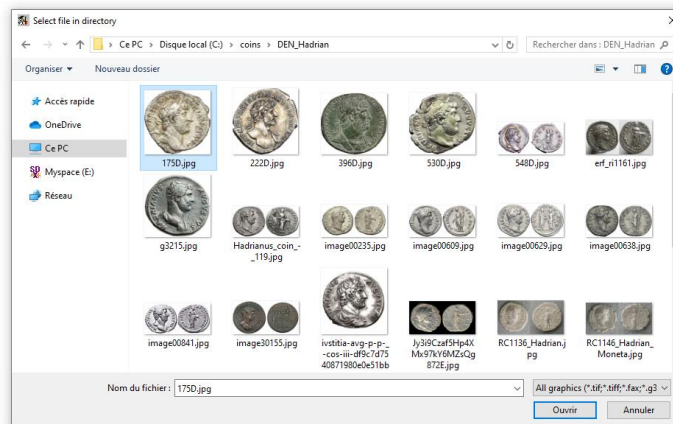
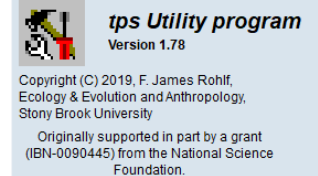
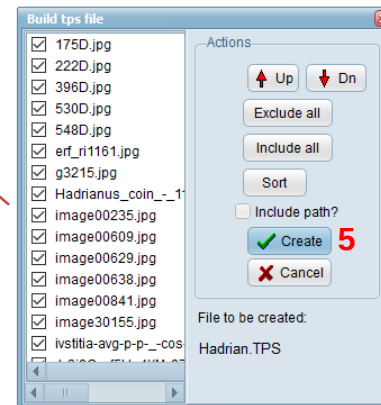
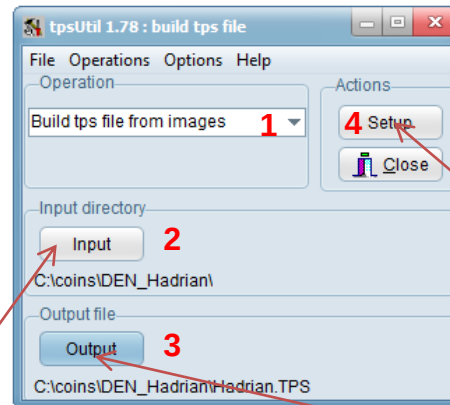
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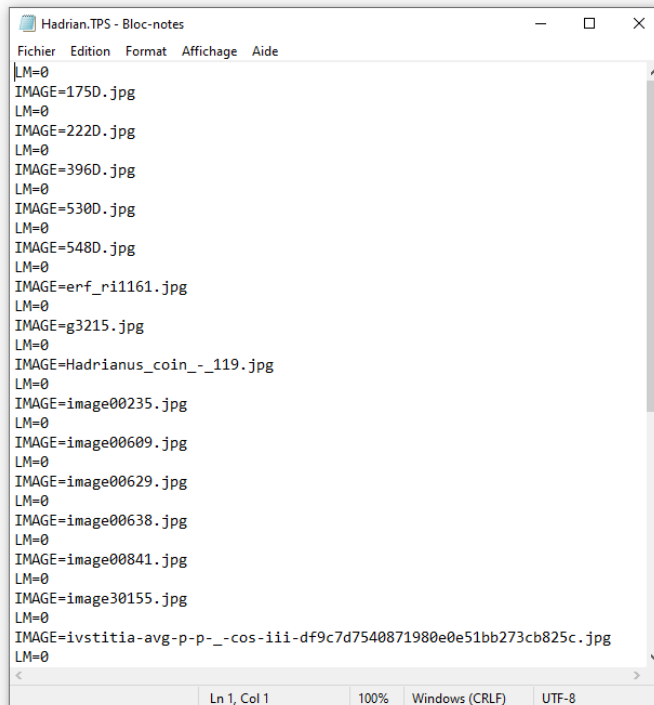
Digitize landmarks

- TPS files
- for both groups of coins

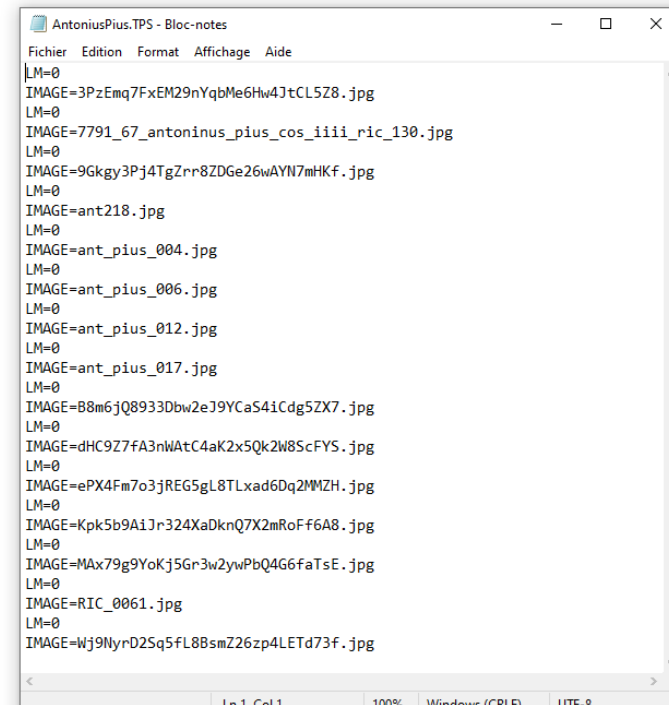


Digitize landmarks

- TPS files



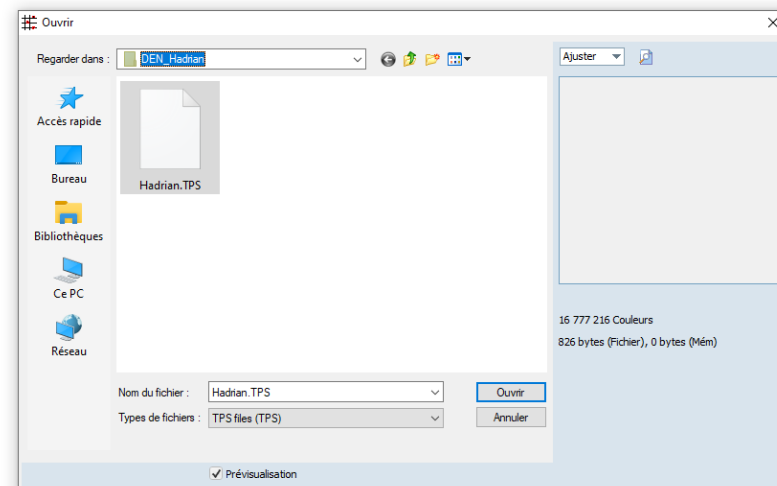
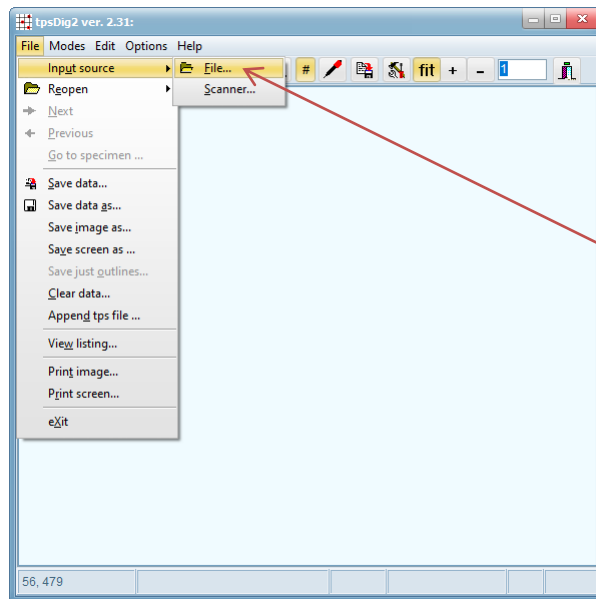
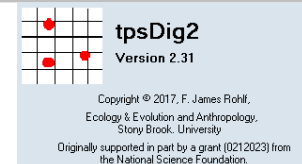
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IMAGE=g3215.jpg
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IMAGE=Hadrianus_coin_-_119.jpg
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LM=0
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```
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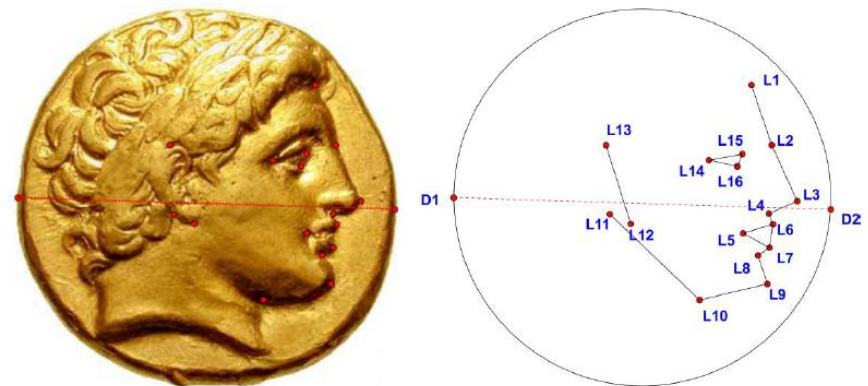
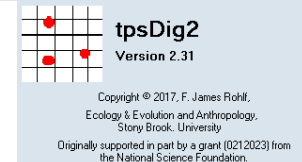
Digitize landmarks

- TPS files



Digitize landmarks

- TPS files



Digitize landmarks

- TPS files

```

Hadrian.TPS - Bloc-notes
Fichier Edition Format Affichage Aide
LM=16
624,00000 690,00000
663,00000 620,00000
729,00000 497,00000
670,00000 480,00000
679,00000 453,00000
642,00000 444,00000
688,00000 419,00000
670,00000 402,00000
680,00000 345,00000
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444,00000 471,00000
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625,00000 622,00000
583,00000 603,00000
619,00000 592,00000
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418,00000 353,00000
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390,00000 277,00000
341,00000 258,00000
248,00000 343,00000
Ln 1, Col 1 100% Windows (CRLF) UTF-8
  
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```

AntoniusPius.TPS - Bloc-notes
Fichier Edition Format Affichage Aide
LM=16
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255,00000 224,00000
241,00000 224,00000
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407,00000 276,00000
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413,00000 217,00000
351,00000 202,00000
237,00000 318,00000
Ln 1, Col 1 100% Windows (CRLF) UTF-8
  
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Make links file

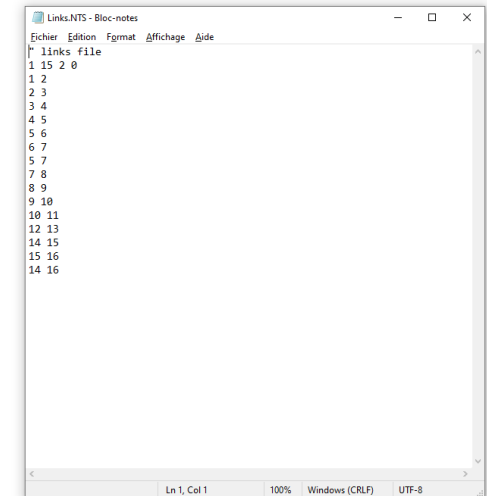
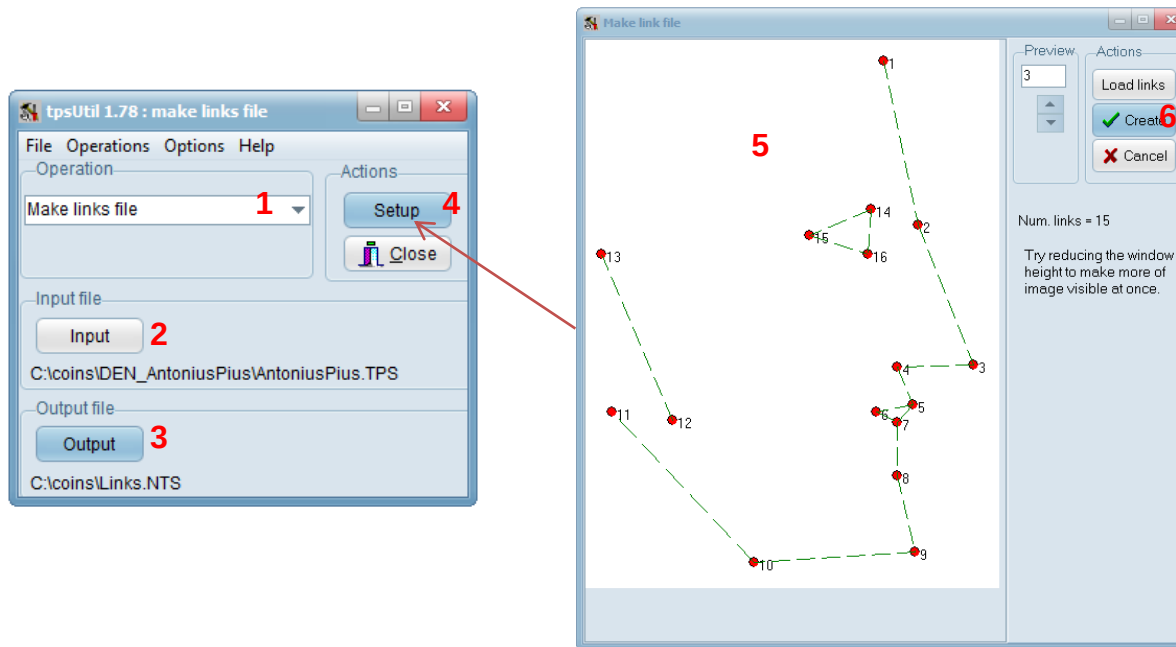
- NTS file



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Version 1.78

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Calculate consensus (meanshapes) for both types of coins

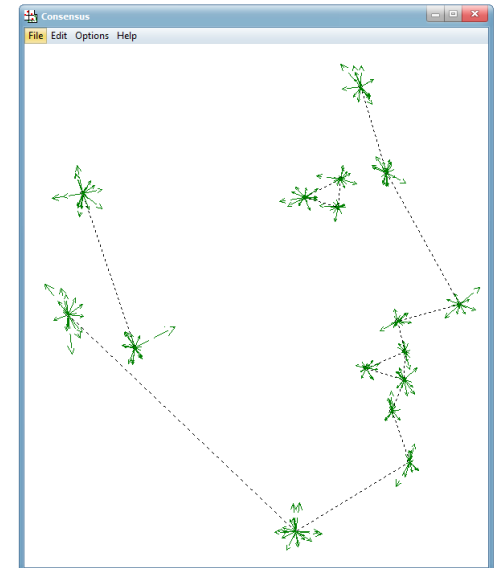
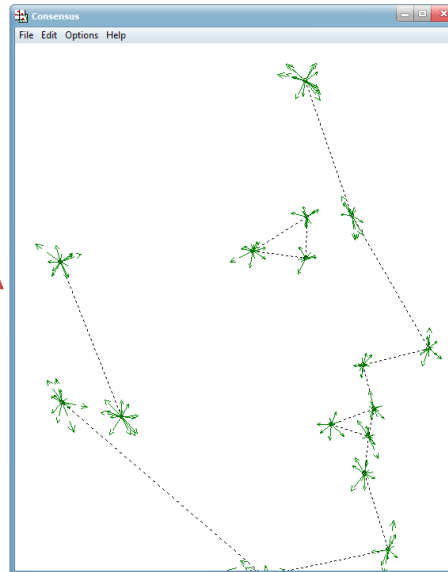
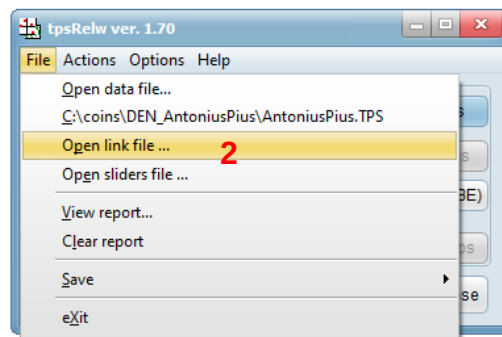
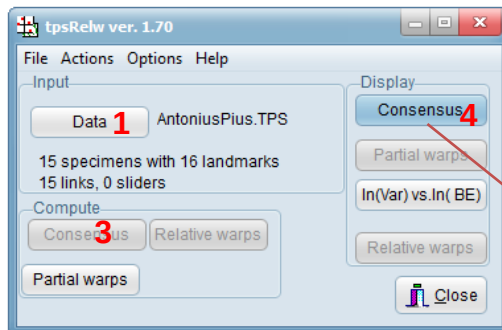
- check differences
- check landmark variations (something strange?)



Relative warps
Version 1.70

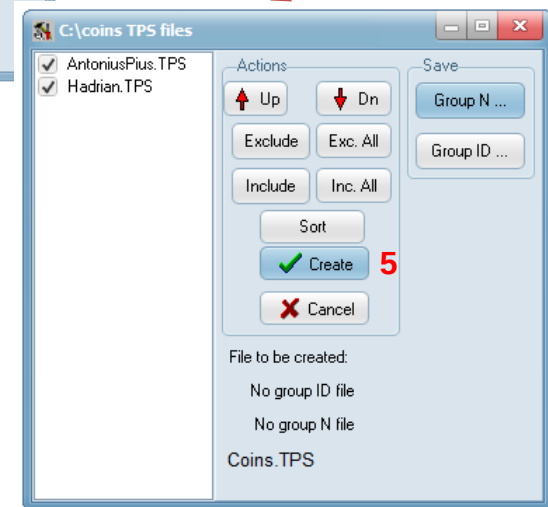
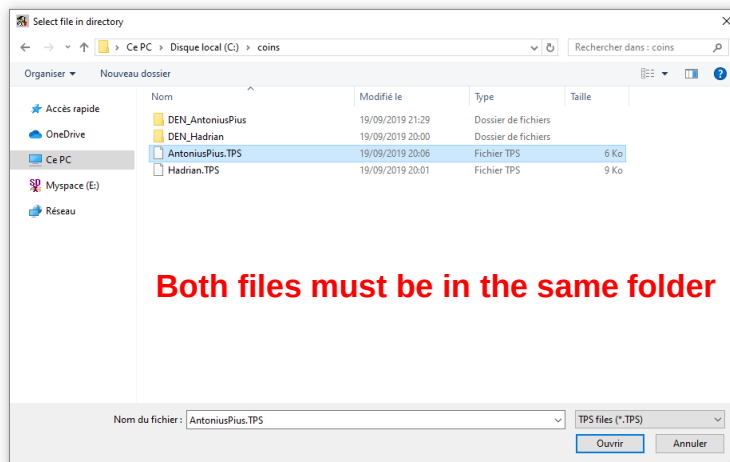
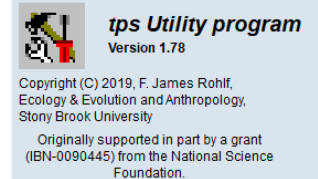
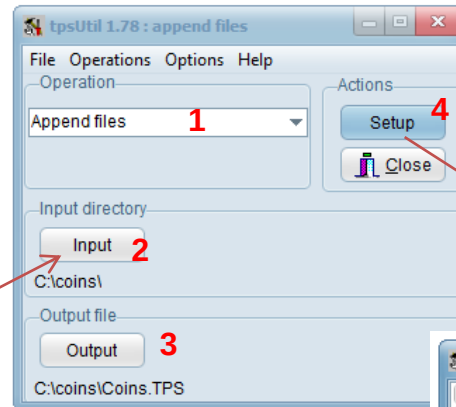
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Merge both TPS files

- Necessary for further analyses



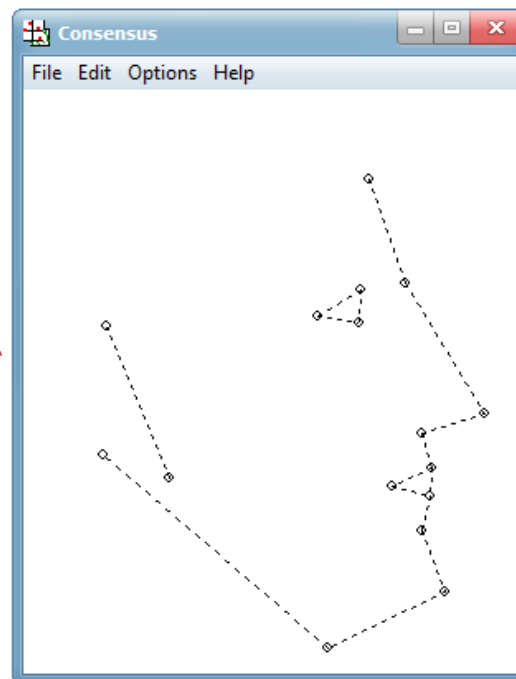
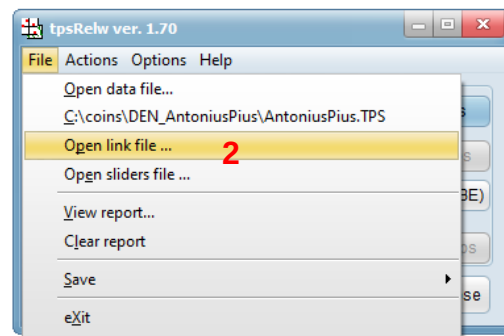
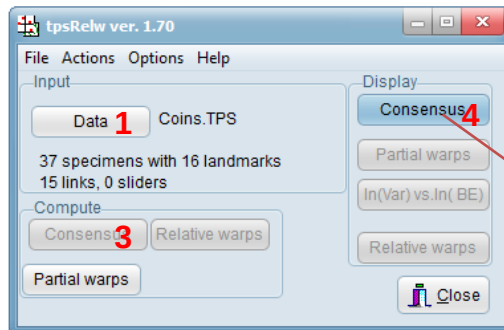
Calculate overall consensus (meanshape)



Relative warps
Version 1.70

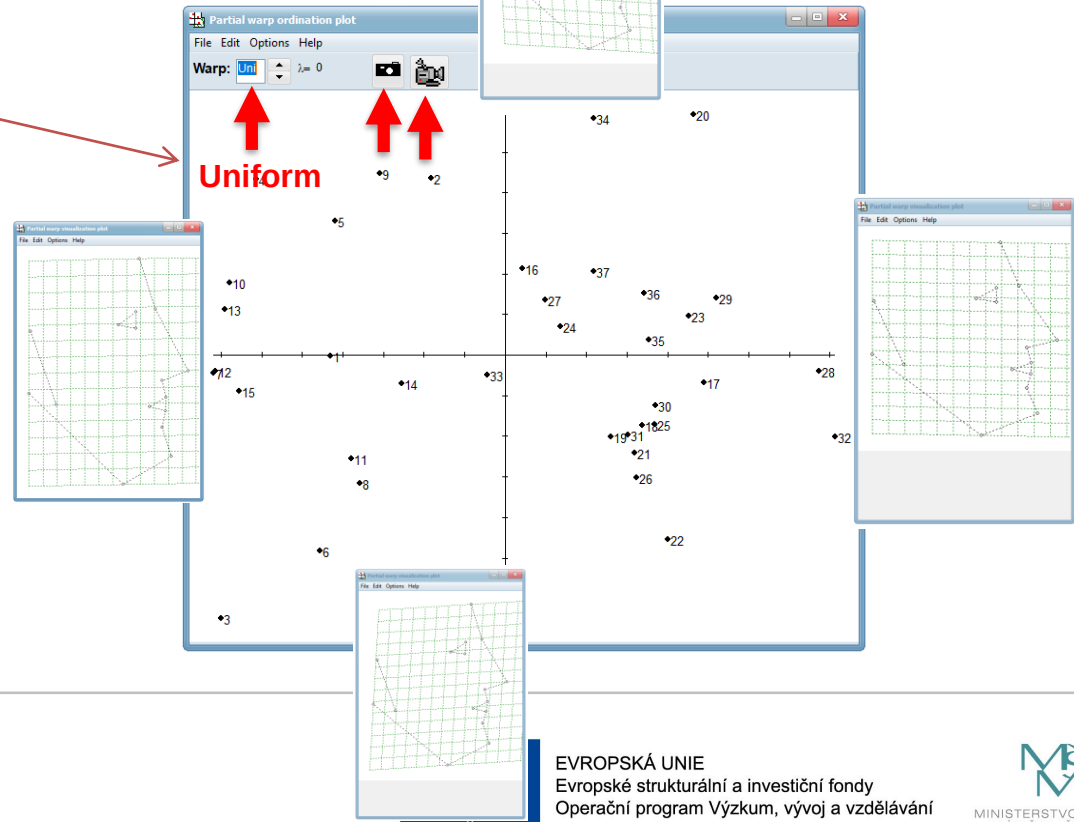
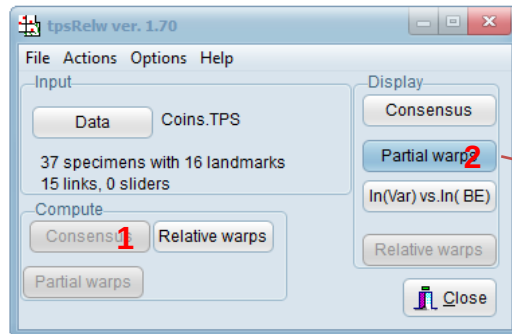
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Partial Warps

Investigation of the uniform component
Description and discussion



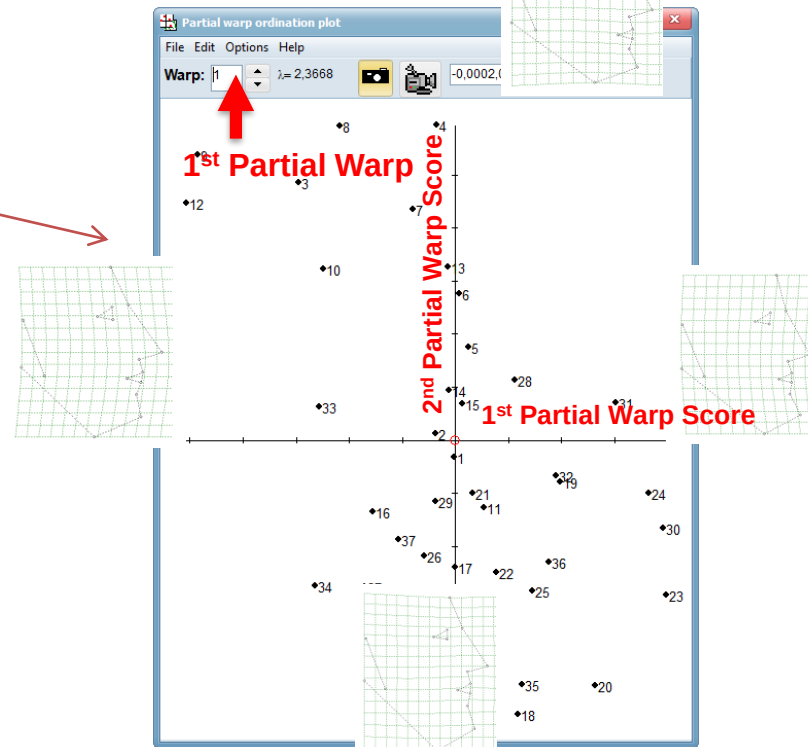
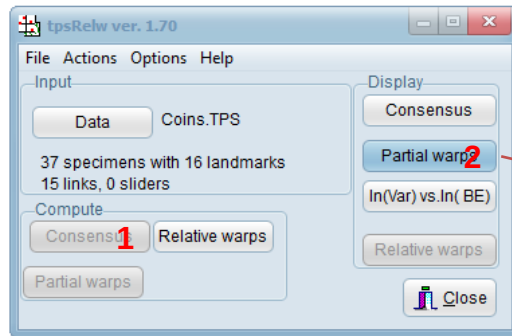
Relative warps
Version 1.70

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Partial Warps

Investigation of the non-uniform component (Partial Warps)
Description and discussion

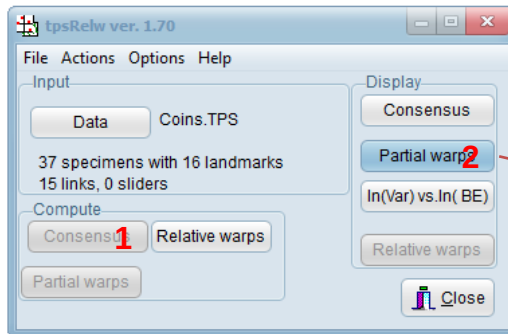


Relative warps
Version 1.70

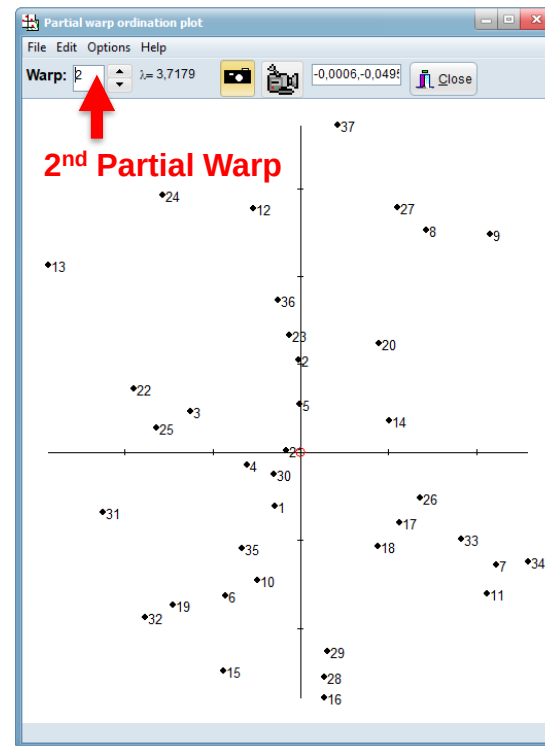
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Partial Warps

Investigation of the non-uniform component (Partial Warps)
Description and discussion



Also warps with lower bending energy...



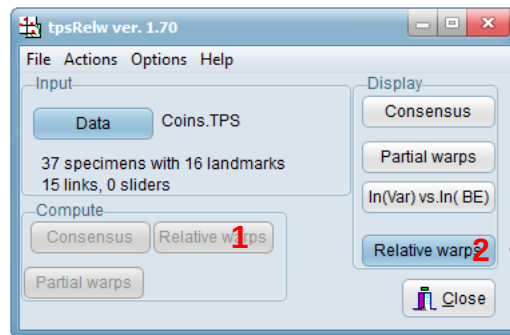
Relative warps
Version 1.70

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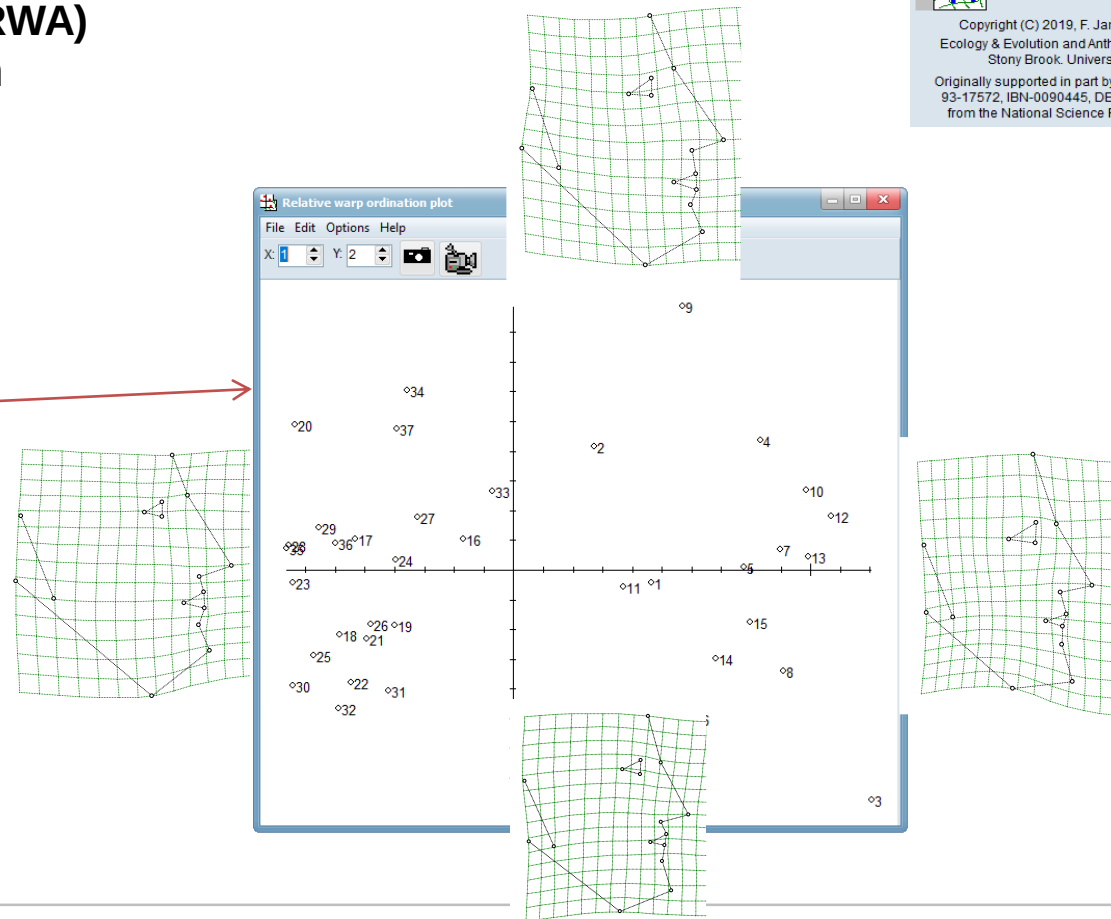
Originally supported in part by grants (DEB
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Relative Warps Analysis (RWA)

Description and discussion



Are there differences?



Relative warps
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from the National Science Foundation.

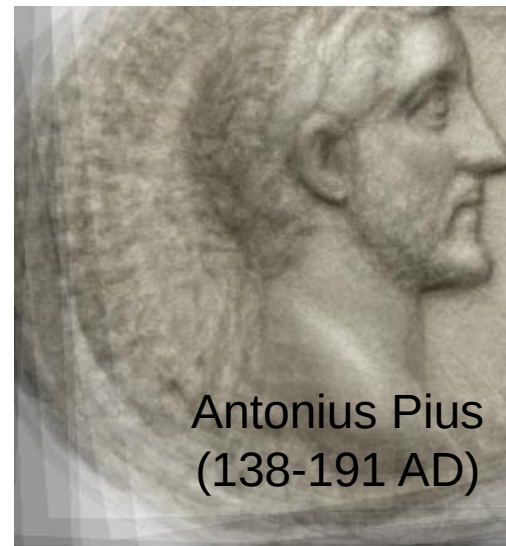
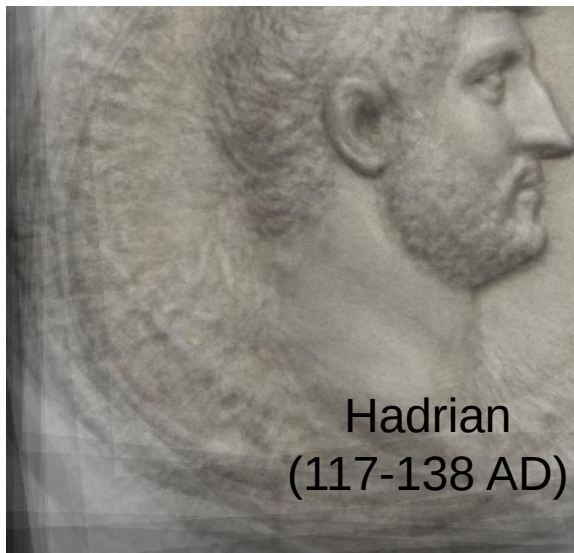
Calculation of bitmap consensus (meanshape)

Optional...



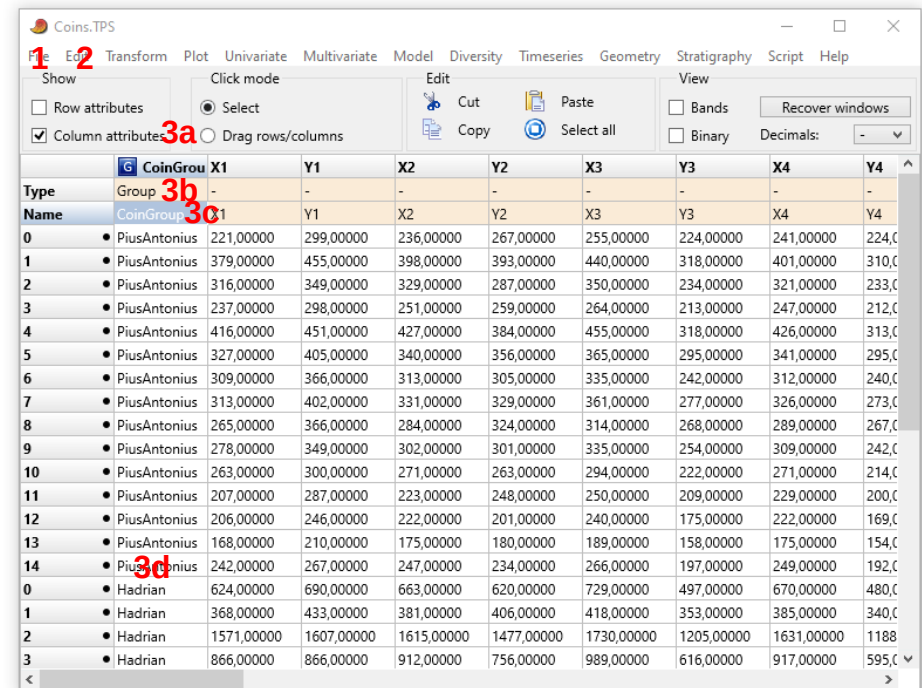
tpsSuper
Version: 2.05

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Foundation.



Data preparation

- 1) Open file:
File >> Open >> Coins.TPS
- 2) Add grouping column for coin type:
Edit >> Insert more columns
- 3) Specify variable type of grouping column and fill it:
 - a) Check Column attributes in Show panel
 - b) Specify Type to Group
 - c) Specify Name of Variable (Optionally)
 - d) Fill group for each coin: 15 x PA, 25 x HAD

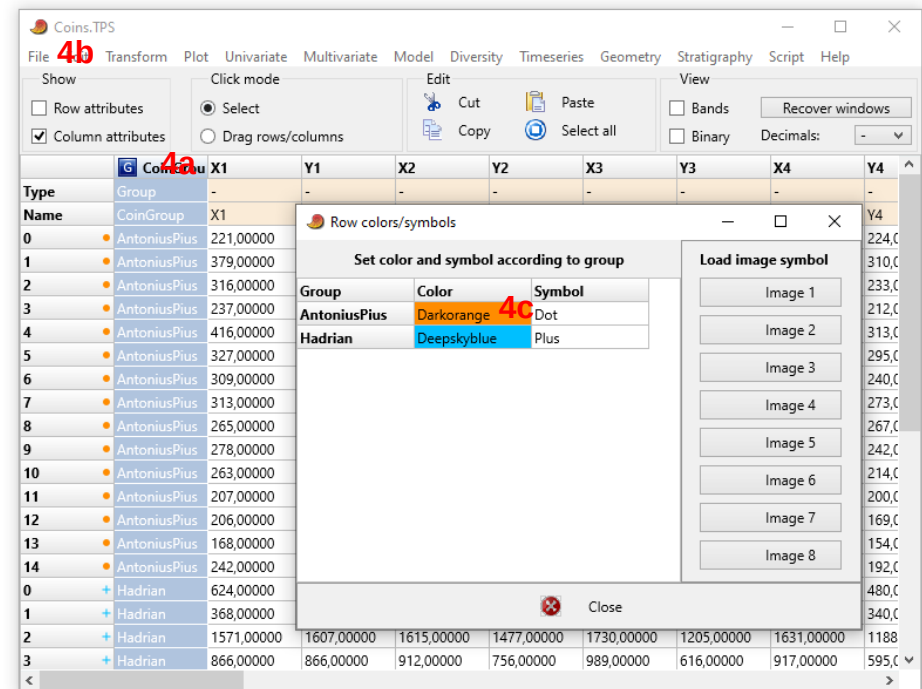


	CoinGroup	X1	Y1	X2	Y2	X3	Y3	X4	Y4
Type	Group	-	-	-	-	-	-	-	-
Name	CoinGroup	X1	Y1	X2	Y2	X3	Y3	X4	Y4
0	• PiusAntonius	221,00000	299,00000	236,00000	267,00000	255,00000	224,00000	241,00000	224,00000
1	• PiusAntonius	379,00000	455,00000	398,00000	393,00000	440,00000	318,00000	401,00000	310,00000
2	• PiusAntonius	316,00000	349,00000	329,00000	287,00000	350,00000	234,00000	321,00000	233,00000
3	• PiusAntonius	237,00000	298,00000	251,00000	259,00000	264,00000	213,00000	247,00000	212,00000
4	• PiusAntonius	416,00000	451,00000	427,00000	384,00000	455,00000	318,00000	426,00000	313,00000
5	• PiusAntonius	327,00000	405,00000	340,00000	356,00000	365,00000	295,00000	341,00000	295,00000
6	• PiusAntonius	309,00000	366,00000	313,00000	305,00000	335,00000	242,00000	312,00000	240,00000
7	• PiusAntonius	313,00000	402,00000	329,00000	361,00000	277,00000	326,00000	273,00000	273,00000
8	• PiusAntonius	265,00000	366,00000	284,00000	324,00000	314,00000	268,00000	289,00000	267,00000
9	• PiusAntonius	278,00000	349,00000	302,00000	301,00000	335,00000	254,00000	309,00000	242,00000
10	• PiusAntonius	263,00000	300,00000	271,00000	263,00000	294,00000	222,00000	271,00000	214,00000
11	• PiusAntonius	207,00000	287,00000	223,00000	248,00000	250,00000	209,00000	229,00000	200,00000
12	• PiusAntonius	206,00000	246,00000	222,00000	201,00000	240,00000	175,00000	222,00000	169,00000
13	• PiusAntonius	168,00000	210,00000	175,00000	180,00000	189,00000	158,00000	175,00000	154,00000
14	• PiusAntonius	242,00000	267,00000	247,00000	234,00000	266,00000	197,00000	249,00000	192,00000
0	• Hadrian	624,00000	690,00000	663,00000	620,00000	729,00000	497,00000	670,00000	480,00000
1	• Hadrian	368,00000	433,00000	381,00000	406,00000	418,00000	353,00000	385,00000	340,00000
2	• Hadrian	1571,00000	1607,00000	1615,00000	1477,00000	1730,00000	1205,00000	1631,00000	1188,00000
3	• Hadrian	866,00000	866,00000	912,00000	756,00000	989,00000	616,00000	917,00000	595,00000

Data preparation



- 1) Open file:
File >> Open >> Coins.TPS
- 2) Add grouping column for coin type:
Edit >> Insert more columns
- 3) Specify variable type of grouping column and fill it:
 - a) Check Column attributes in Show panel
 - b) Specify Type to Group
 - c) Specify Name of Variable (Optionally)
 - d) Fill group for each coin: 15 x PA, 25 x HAD
- 4) Specify colours for coins based on grouping column
 - a) Select grouping column
 - b) Edit >> Row colors/symbols
 - c) Adjust as you wish



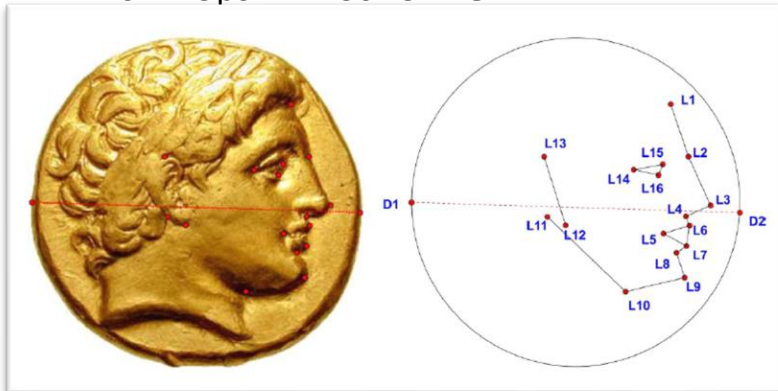
The screenshot shows the 'Coins.TPS' software interface. The main window displays a table with columns for coin type, value, and coordinates. A 'Row colors/symbols' dialog box is open, allowing users to set colors and symbols for different groups (AntoniusPius, Hadrian). The dialog shows 'AntoniusPius' with a 'Darkorange' color and 'Hadrian' with a 'Deepskyblue' color. The 'Symbol' column is set to 'Dot' for AntoniusPius and 'Plus' for Hadrian.

Type	Group	X1	Y1	X2	Y2	X3	Y3	X4	Y4
0	AntoniusPius	221,00000							
1	AntoniusPius	379,00000							
2	AntoniusPius	316,00000							
3	AntoniusPius	237,00000							
4	AntoniusPius	416,00000							
5	AntoniusPius	327,00000							
6	AntoniusPius	309,00000							
7	AntoniusPius	313,00000							
8	AntoniusPius	265,00000							
9	AntoniusPius	278,00000							
10	AntoniusPius	263,00000							
11	AntoniusPius	207,00000							
12	AntoniusPius	206,00000							
13	AntoniusPius	168,00000							
14	AntoniusPius	242,00000							
0	Hadrian	624,00000							
1	Hadrian	368,00000							
2	Hadrian	1571,00000	1607,00000	1615,00000	1477,00000	1730,00000	1205,00000	1631,00000	1188
3	Hadrian	866,00000	866,00000	912,00000	756,00000	989,00000	616,00000	917,00000	595,00000

Data preparation

1) Open file:

File >> Open >> Coins.TPS

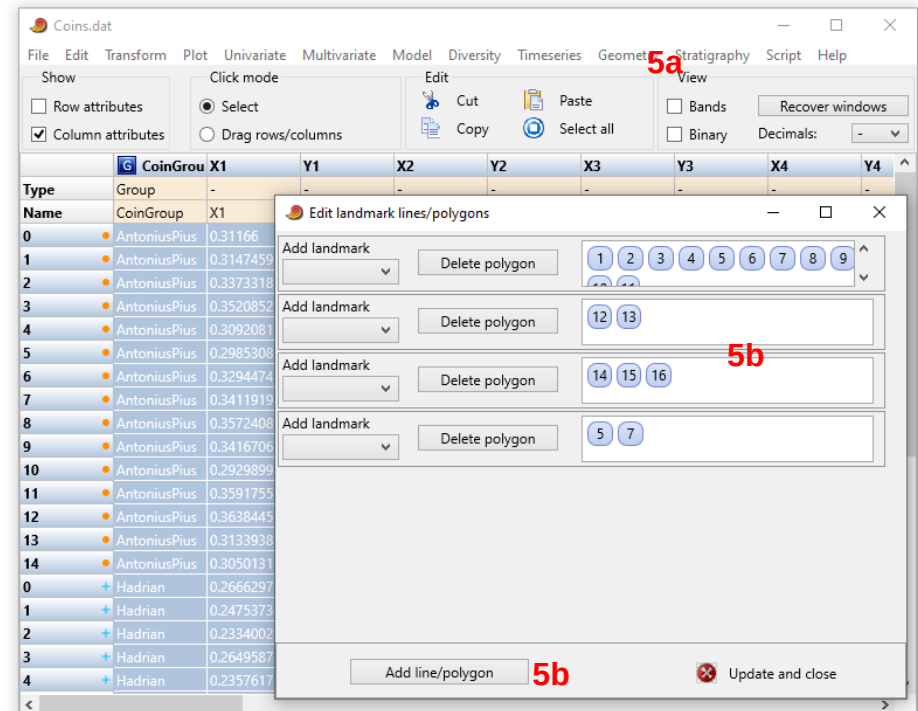


4) Specify colours for coins based on grouping column

- Select grouping column
- Edit >> Row colors/symbols
- Adjust as you wish

5) Add links

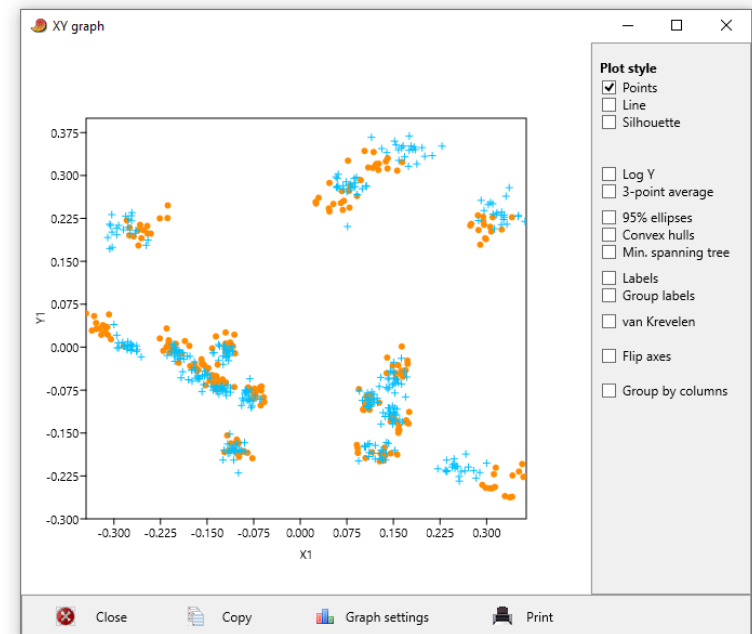
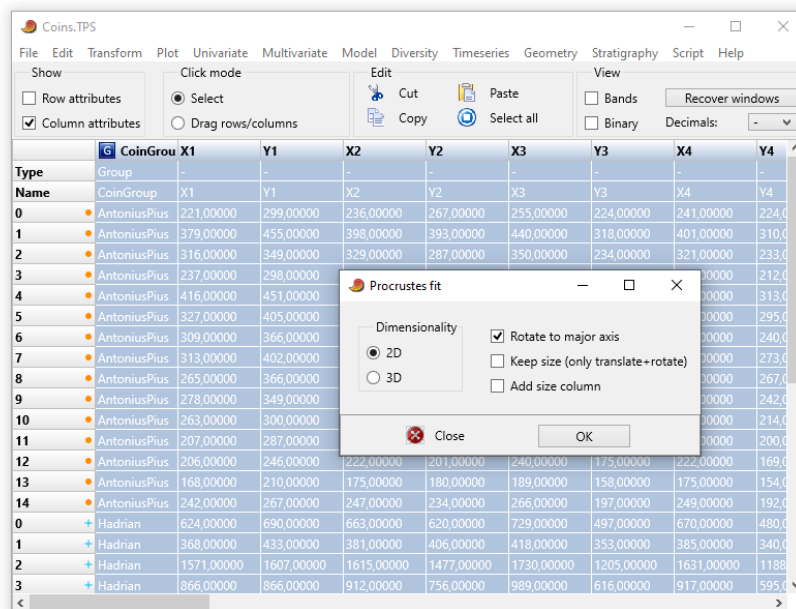
- Geometry >> Landmarks (2D) >> Edit landmark lines/polygons
- Specify links



Procrustes Analysis

Transform >> Landmarks >> Procrustes

Plot >> XY graph

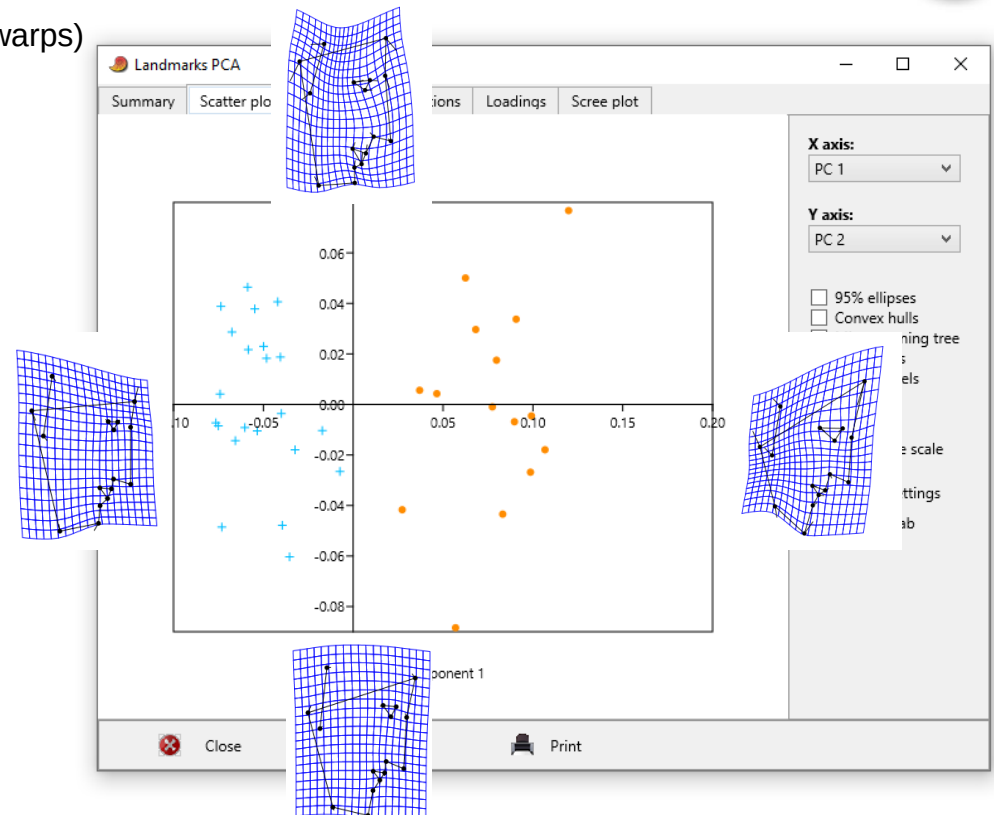


Relative warps Analysis

Geometry >> Landmarks (2D) >> PCA (relative warps)

Description and discussion

- how much information we see in graphics?
- how many components are meaningful?
- which changes of shape are represented by relative warps?
- are there differences between groups?
- what characterise them in the shape
- which group has more shape variability?
- are there outliers? If so, comment.
- ...



Relative warps Analysis

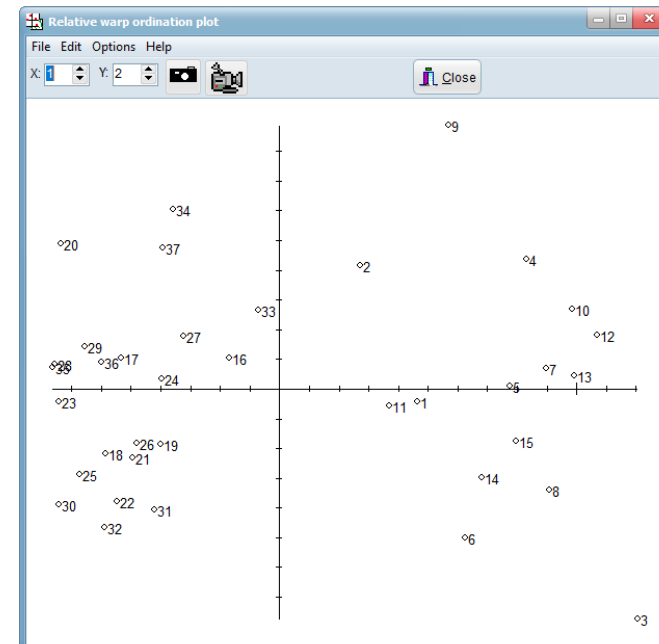
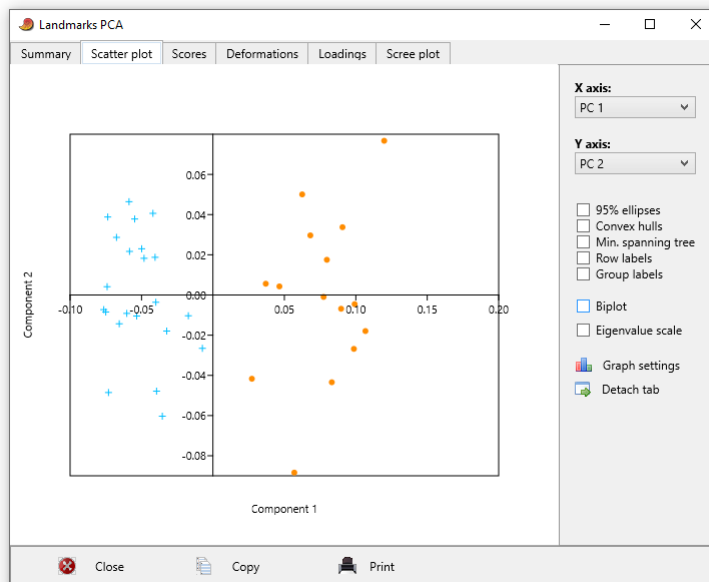
check if PAST and tpsrelw32 show the same thing.



Relative warps
Version 1.70

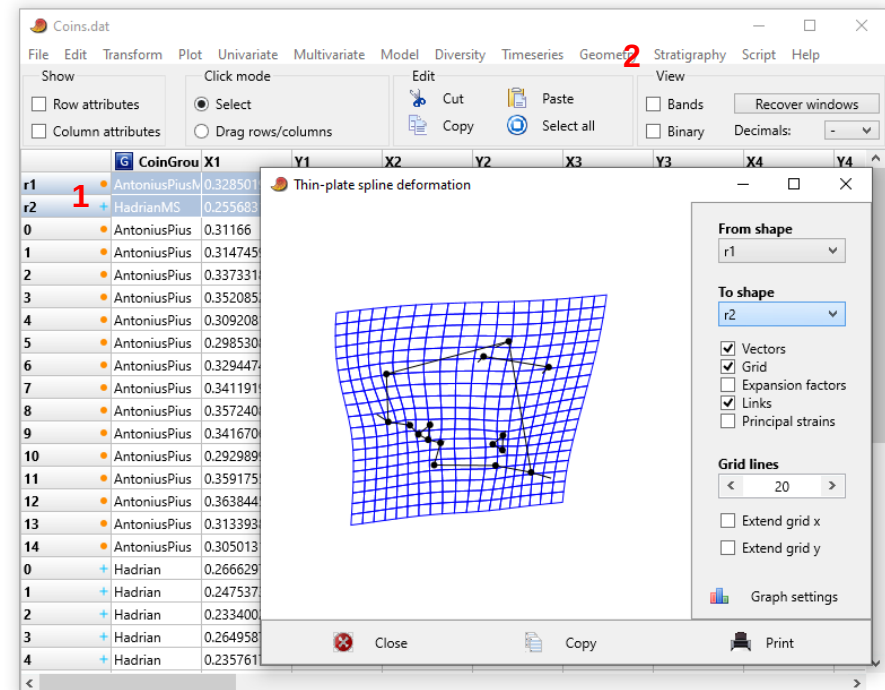
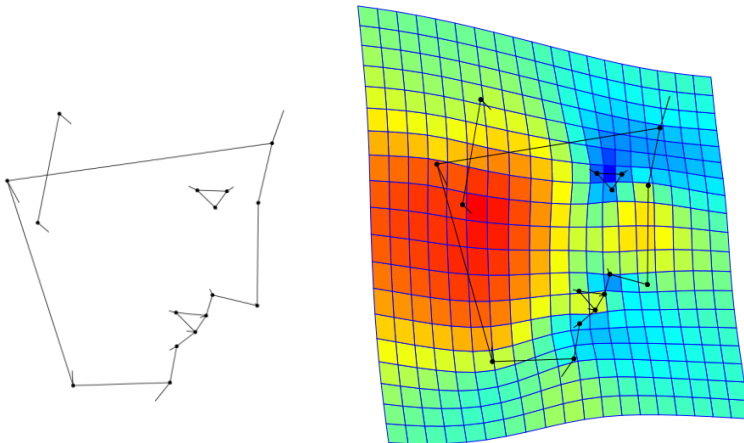
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Thin-plate splines (TPS) Comparison of group meanshapes

- 1) Calculate mean coordinates for both groups externally (e.g. in Excel) and import results to PAST.
- if complicated, import "Mean_shape.xlsx"
- 2) Geometry >> Landmarks (2D) >> Thin-plate splines



References

- <https://jwilczek.com/teaching/m2-ages-morphometrie/>
- <http://www.fabricemonna.com/enseignement-2/master-ages/morphometrie-m2-ages/>

